
MSCA Digital Finance

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Preliminaries on Stochastic Processes and Stochastic Calculus

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Why these preliminaries?

- ▶ The modelling of stock prices and investment strategies will be seen as random experiments that evolve over time and that are depending on the past. We thus need *stochastic processes*.
- ▶ Investment strategies should not incorporate knowledge of the future.
- ▶ We will need the Itô-integral for modelling stock prices, a technical tool that is typically not taught in probability courses.
- ▶ Digital assets have an even more irregular behaviour than physical assets. As a consequence we need an extension of the Itô-integral to include jump processes.
- ▶ Being able to work with Itô-integrals, Brownian motion and Poisson processes is a standard requirement on mathematicians working in the finance and modern insurance industry.

Hence: We will need at least a basic knowledge in these topics.

Stochastic Processes – Definition and Properties

Definition

- a) A (d-dimensional) **stochastic process** X is a collection of ($\mathbf{R}^d$ -valued) random variables $\{X_t, t \in I\}$ where I is an interval contained in the non-negative real numbers. For fixed $\omega \in \Omega$, the set $\{X_t(\omega), t \in I\}$ is called a **path** (or a realisation) of X .
- b) A **filtration** $\{F_t, t \in [0, T]\}$ is a non-decreasing sequence of sub- σ -fields of F . The quadruple $(\Omega, F, P, \{F_t, t \in [0, T]\})$ will be called a **filtered probability space**.
- c) A filtration $\{F_t, t \in [0, T]\}$ is called **right-continuous** (**left-continuous**) if it satisfies

$$F_t = F_{t+} := \bigcap_{\varepsilon > 0} F_{t+\varepsilon} \forall 0 \leq t < T, \left(F_t = F_{t-} := \sigma \left(\bigcup_{s < t} F_s \right) \right) \quad (1)$$

It is called *continuous* if it is both left- and right-continuous.

Stochastic Processes – Definition and Properties

Definition (Continuation)

d) We call a stochastic process X *adapted to the filtration* $\{F_t, t \in [0, T]\}$ if for every t X_t is F_t measurable. In this case, we use the notation $\{(X_t, F_t), t \in [0, T]\}$ for the stochastic process.

Definition

Let $\{(X_t, F_t), t \in [0, T]\}$ be a real valued stochastic process with $E(|X_t|) < \infty$ for all $t \in [0, T]$.

a) $\{(X_t, F_t), t \in [0, T]\}$ is called a **martingale** if we have

$$E(X_t | F_s) = X_s \quad \forall 0 \leq s \leq t \leq T. \quad (2)$$

b) It is called a **submartingale** (supermartingale) if we have

$$E(X_t | F_s) \geq X_s \quad (E(X_t | F_s) \leq X_s) \quad \forall 0 \leq s \leq t \leq T. \quad (3)$$

Brownian Motion – Motivation

Itô calculus compact

Definition

a) A real valued stochastic process $\{W_t\}_{t \geq 0}$ with continuous paths and

1. $W_0 = 0$ P -almost surely
2. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$, *stationary increments*
3. $W_t - W_s$ independent of $W_u - W_r$ for $0 \leq r \leq u \leq s < t$,
independent increments

is called a *one dimensional Brownian Motion*.

b) An $\mathbb{R}^n$ -valued process of independent one dim. Brown. Motions W_i

$$W(t) = (W_1(t), \dots, W_n(t)),$$

is called an *n-dimensional Brownian Motion*.

Note: A Brownian motion is a martingale.

Brownian Motion – Simulation

It is extremely simple to simulate a (discretized and interpolated) Brownian Motion!

What you need for simulation of a Br. motion on $[0, T]$:

- the choice of a discretization $\Delta = T/N$,
- a method to generate independent $N(0, 1)$ -random numbers Z_i

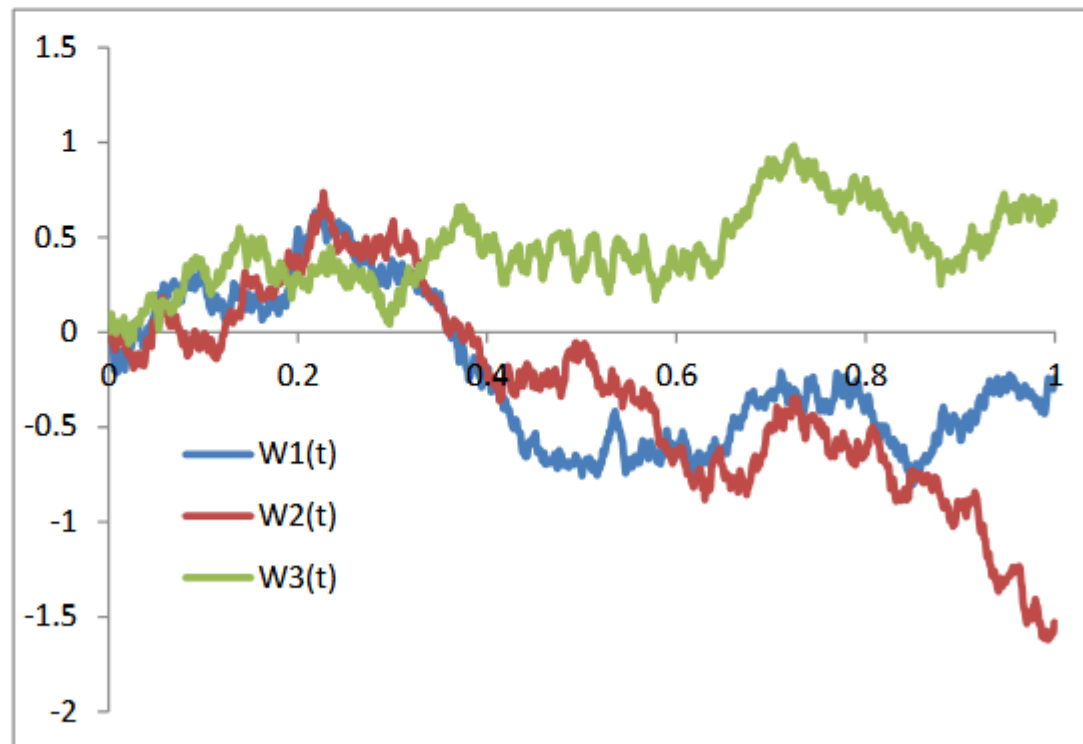
Algorithm to generate (a path of) a Brownian motion:

- Set $W(0) = 0$.
- For $i = 1$ to N :
 - ▶ Generate an $N(0, 1)$ distributed random number Z_i .
 - ▶ Set $W(i\Delta) = W((i-1)\Delta) + \sqrt{\Delta}Z_i$.
 - ▶ Linearly interpolate between $W((i-1)\Delta)$ and $W(i\Delta)$.

Brownian Motion – Simulation

It is extremely simple to simulate a (discretized and interpolated) Brownian Motion!

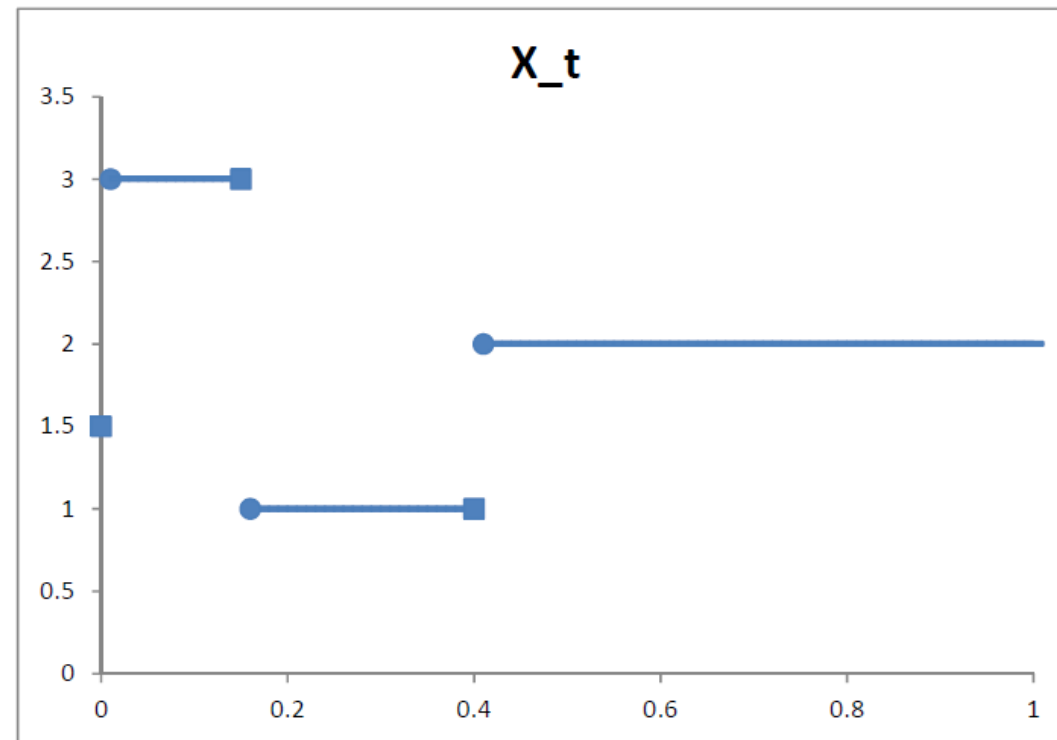
Three simulated paths of a one-dim. Br. motion



The Itô-Calculus as tool

Stochastic integral for simple processes

Path of a simple process



The Itô-Calculus as tool

Stochastic integral for simple processes – 2

Definition

For a simple process $\{X_t\}_{t \in [0, T]}$, we define the *stochastic integral* $I_t(X)$ for $t \in (t_k, t_{k+1}]$ as

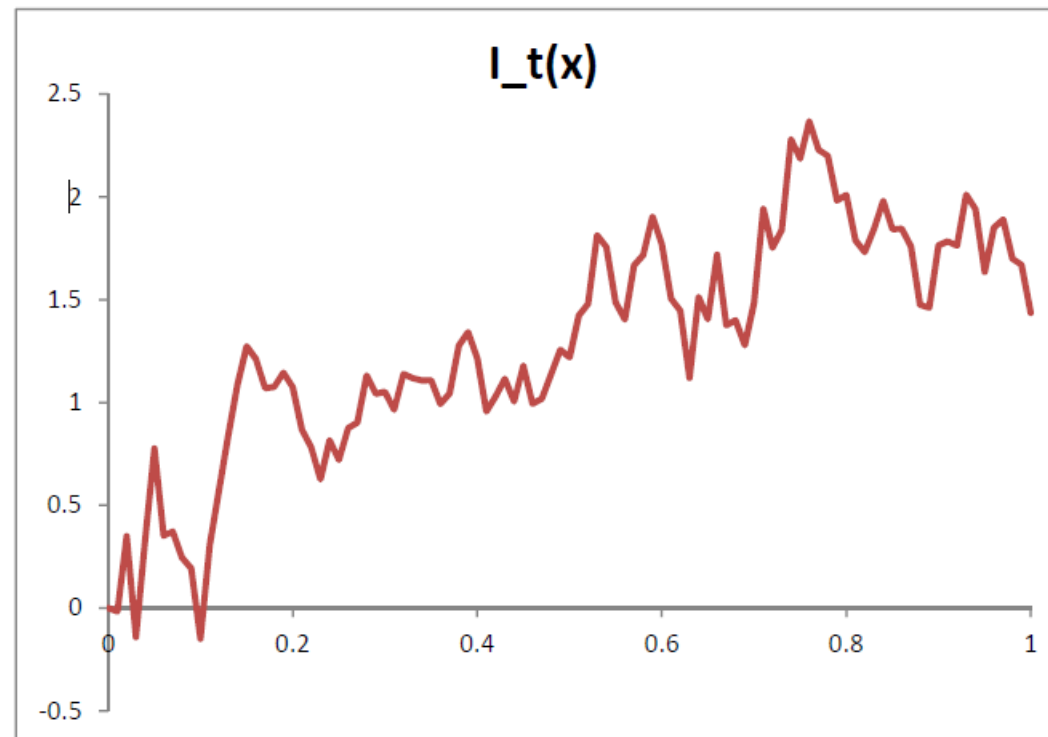
$$I_t(X) := \int_0^t X_s dW_s := \sum_{1 \leq i \leq k} \Phi_i (W_{t_i} - W_{t_{i-1}}) + \Phi_{k+1} (W_t - W_{t_k}) \quad (1)$$

Note: We multiply the increments of the Brown. motion on an interval on which X is constant by the values of X_t , namely Φ_i (compare to the Lebesgue-Stieltjes-integral for simple functions).

The Itô-Calculus as tool

Stochastic integral for simple processes – 3

Stoch. integral of a simple process (Why does it have this form?)



The Itô-Calculus as tool

Stochastic integral for simple processes – 4

Theorem (Properties of the stochastic integral)

$X := \{X_t\}_{t \in [0, T]}$ a simple process. Then we have:

a) $\{(I_t(X))\}_{t \in [0, T]}$ is a continuous martingale w.r.t. $\{F_t\}_{t \in [0, T]}$. In particular we have $E(I_t(X)) = 0$ for all $t \in [0, T]$.

$$b) E \left(\int_0^t X_s dW_s \right)^2 = E \left(\int_0^t X_s^2 ds \right) \text{ for } t \in [0, T].$$

c) $I_t(aX + bY) = aI_t(X) + bI_t(Y)$ for simple processes X, Y , $a, b \in \mathbb{R}$.

Rem.: The definition of the stoch. integral will be extended to more general processes $X \in L^2[0, T]$ for which the right hand side of Equation b) is finite.

Working with stochastic integrals

Itô-processes

Definition

$\{(W(t), F_t)\}_{t \in [0, \infty)}$ a Brownian motion. $\{(X(t), F_t)\}_{t \in [0, \infty)}$ is called *real valued Itô-process*, if for all $t \geq 0$ it has a representation of the form

$$X(t) = X(0) + \int_0^t K(s) ds + \int_0^t H(s) dW(s) \quad P\text{-a.s.}$$

where $X(0)$ F_0 -measurable and $\{K(t)\}_{t \in [0, \infty)}$, $\{H(t)\}_{t \in [0, \infty)}$ are progressively measurable, *sufficiently integrable* processes.

Working with Itô-processes

The classical Itô-Formula

Theorem (One-dimensional Itô-Formula)

W_t 1-dim. Brownian motion, X_t real valued Itô-process with

$$X_t = X_0 + \int_0^t K_s ds + \int_0^t H_s dW_s, f : \mathbb{R} \rightarrow \mathbb{R} \in C^2.$$

Then, we have for all $t \geq 0$:

$$f(X_t) = f(X_0) + \int_0^t \left(f'(X_s) K_s + \frac{1}{2} f''(X_s) H_s^2 \right) ds + \int_0^t f'(X_s) H_s dW_s \quad P\text{-f.s.} \quad (2)$$

In particular all appearing integrals are defined and $f(X_t)$ is again an Itô-Process.

Working with Itô-processes

The product rule

Theorem (Product rule)

X_t and Y_t one dimensional Itô-Processes with

$$X_t = X_0 + \int_0^t K_s ds + \int_0^t H_s dW_s, Y_t = Y_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s.$$

Then we have:

$$X_t Y_t = X_0 Y_0 + \int_0^t (X_s \mu_s + Y_s K_s + H_s \sigma_s) ds + \int_0^t (X_s \sigma_s + Y_s H_s) dW_s .$$

Rem.: $\langle X, Y \rangle_t := \int_0^t H_s \sigma_s ds$ is called the quadratic covariation of X, Y , $\langle X \rangle_t = \langle X, X \rangle_t$ is called the quadratic variation of X .

Working with Itô-processes

The product rule – 2

With the (formal) differential representation

$$dX_t = K_t dt + H_t dW_t, dY_t = \mu_t ds + \sigma_t dW_t$$

the product rule is given as

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d \langle X, Y \rangle_t .$$

We also have a time-dependent Itô-Formula in differential notation

$$df(t, X_t) = f_t(t, X_t)dt + f_x(t, X_t)dX_t + \frac{1}{2}f_{xx}(t, X_t)d \langle X \rangle_t$$

and many more variants.

Working with Itô-processes

Examples: Using the Itô-Formula

Example 1 $X_t = W_t = 0 + \int_0^t 0 \, ds + \int_0^t 1 \, dW_s, \, f(x) = x^2$

$$W_t^2 = 0^2 + \int_0^t \left(2 * 0 + \frac{1}{2}2 * 1\right) ds + \int_0^t 2W_s * 1 \, dW_s = t + \int_0^t 2W_s \, dW_s$$

This formula shows the difference of the Itô formula to the usual chain rule, the additional term t . It has its origin in the fact that the Brown. motion has a non-vanishing quadratic variation of $\langle W \rangle_t = t$.

Working with Itô-processes

Equation of the stock price

Example 2 For $X_t = (b - \frac{1}{2}\sigma^2)t + \sigma W_t$ using the Itô formula with $f(x) = S_0 e^x$, i.e. for $S_t = S_0 e^{(b - \frac{1}{2}\sigma^2)t + \sigma W_t} = f(X_t)$, we obtain:

$$\begin{aligned} S_t &= S_0 + \int_0^t (S_s(b - \frac{1}{2}\sigma^2) + \frac{1}{2}S_s\sigma^2) ds + \int_0^t S_s\sigma dW_s \\ &= S_0 + \int_0^t S_s b ds + \int_0^t S_s\sigma dW_s \end{aligned}$$

This integral equation in differential notation

$$(SDE) \quad dS_t = S_t (bdt + \sigma dW_t)$$

is called a *stochastic differential equation*. One can show that the above stock price $S_t = S_0 e^{(b - \frac{1}{2}\sigma^2)t + \sigma W_t}$ is the unique solution of this SDE with initial value S_0 .

Working with Itô-processes

Variation of constants

Theorem (Variation of constants, simple version)

$$dX(t) = (A(t)X(t) + a(t))dt + (S(t)X(t) + \sigma(t))dW(t), \quad X(0) = x$$

$$\Rightarrow X(t) = Z(t) \cdot \left(x + \int_0^t \frac{1}{Z(u)} \left(a(u) - S(u)\sigma(u) \right) du + \int_0^t \frac{\sigma(u)}{Z(u)} dW(u) \right)$$

with

$$Z(t) = \exp \left(\int_0^t \left(A(u) - \frac{1}{2} \cdot \|S(u)\|^2 \right) du + \int_0^t S(u) dW(u) \right)$$

Working with Itô-processes

Variation of constants – 2

In addition:

- ▶ Note that $Z(t)$ in the Theorem is the unique solution of the homogeneous equation

$$dZ(t) = Z(t) (A(t) dt + S(t) dW(t)), \quad Z(0) = 1$$

- ▶ The theorem is also valid for an m -dimensional Brown. motion if $S(t), \sigma(t)$ $\mathbb{R}^m$ -valued stochastic processes which are square-integrable and progressively measurable.
- ▶ The processes $A(t), a(t)$ should be integrable and progressively measurable.
- ▶ The Variation of Constants Theorem already covers many important equations for applications in financial mathematics.